NPH INSULIN

High-alert drug

Cloudy — roll before use

CHEMICAL NAME Human insulin complexed with protamine sulfate and zinc (neutral protamine Hagedorn suspension)

CLASS Intermediate-acting insulin

TRADE NAMES Insulatard · Humulin N / Humulin I · Insuman Basal · Novolin N

THERAPEUTIC USES / INDICATIONS

- Basal insulin in type 1 and type 2 diabetes (once or twice daily).
- Gestational diabetes and diabetes in pregnancy (well-established safety).
- Steroid-induced hyperglycaemia (morning dose).

PHARMACODYNAMICS

- Same as human insulin: ↑ peripheral glucose uptake, ↓ hepatic glucose production.
- Peak action can cause nocturnal hypoglycaemia.

ADMINISTRATION

- **SC only** — never IV.
- Cloudy suspension: gently roll and invert 10–20 times until uniformly milky; do not shake vigorously.
- Often at bedtime and/or before breakfast; thigh for slower absorption at night.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Injection-site reactions, lipohypertrophy.
- Weight gain.
- Oedema (early).

ADVERSE EFFECTS (SERIOUS)

- Hypoglycaemia (especially night-time, around peak).
- Hypokalaemia.
- Allergy to protamine (rare).

DOSAGES

- **Start (type 2):** individualised, often once daily at bedtime, titrated to fasting glucose; usual requirement 0.3–1.0 unit/kg/day (all insulin).
- **Type 1:** part of basal-bolus (basal insulin morning and/or evening with mealtime insulin).
- **Steroid hyperglycaemia:** morning NPH given with once-daily prednisolone, dose per local protocol.

CONTRAINDICATIONS & CAUTIONS

- Hypoglycaemia.
- IV administration.
- Hypersensitivity.

KEY INTERACTIONS

- Same as other insulins: β-blockers mask hypos; steroids ↑ requirements; alcohol ↑ hypoglycaemia.

PHARMACOKINETICS

- **Onset:** 1–2 h. **Peak:** 4–8 h. **Duration:** 12–18 h.
- Protamine delays absorption.
- Metabolised in liver/kidney.

NURSING MANAGEMENT

- Roll vial/pen before use; check uniform cloudiness.
- When mixing with regular insulin in a syringe: draw clear before cloudy.
- Monitor fasting and bedtime glucose; check 3 am glucose if nocturnal hypos suspected.
- Bedtime snack may be needed.

PATIENT EDUCATION

- Roll the pen gently before each injection.
- Recognise night-time hypos (sweats, nightmares, morning headache).
- Have a bedtime snack if advised.
- Rotate injection sites.

KEY POINTS

- Intermediate-acting basal insulin; peak 4–8 h.
- Clear before cloudy when mixing.
- Cloudy — roll, don't shake.
- Nocturnal hypoglycaemia risk.

SOURCES CHECKED **[589]** Insulatard 100 international units/ml suspension for injection in vial (UK SmPC) **[588]** Umpierrez et al., Diabetologia 2024 *Full references p. 432–457*